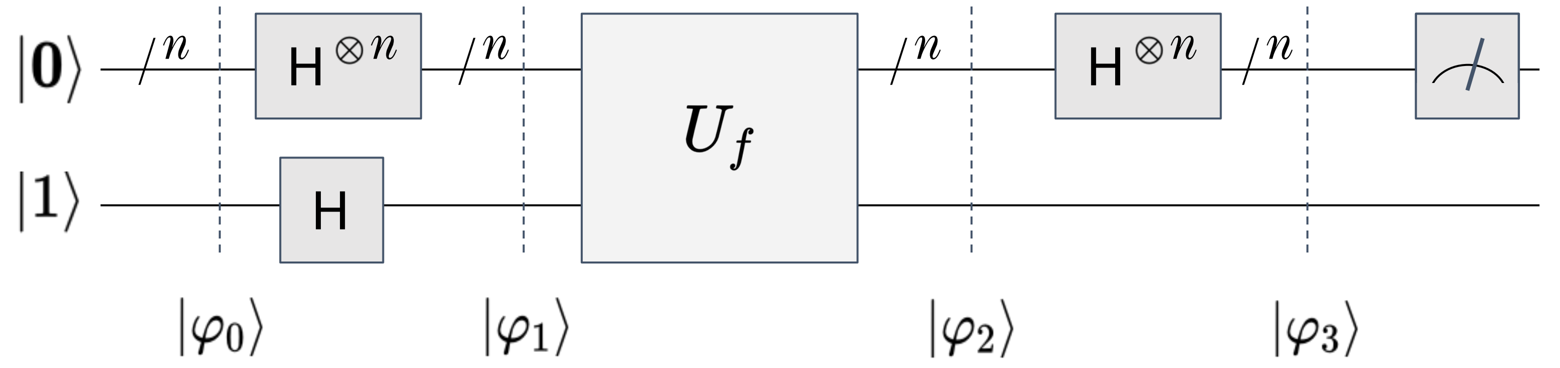
Bernstein-Vazirani Algorithm

$f \longrightarrow s\cdot x \pmod 2$



$|0\rangle^{\otimes n}$ — U_f

$|-\rangle$

$$\sum_{x\in\{0,1\}^n} \underbrace{(-1)^{s\cdot x}|x\rangle}_{(-1)^{f(x)}|x\rangle} = (|0\rangle + (-1)^{s_1}|1\rangle)\otimes(|0\rangle + (-1)^{s_2}|1\rangle)\otimes\cdots\otimes(|0\rangle + (-1)^{s_n}|1\rangle)$$

$(|0\rangle + (-1)^{s_j}|1\rangle)$

$s = s_1 s_2 s_3 \dots s_n$

$s_j = 0$: $\varphi_j = |0\rangle + |1\rangle = |+\rangle$

$s_j = 1$: $\varphi_j = |0\rangle - |1\rangle = |-\rangle$

The expression for $|\varphi_3\rangle$ is independent of the definition of f. We ask the question, for what choices of f could measuring $|\varphi_3\rangle$ yield interesting information?

$$|\varphi_3\rangle = \frac{1}{2^n}\sum_{z\in\{0,1\}^n}\left(\sum_{x\in\{0,1\}^n}(-1)^{f(x)+x\cdot z}\right)|z\rangle|-\rangle$$

$f(x) = s\cdot x \bmod 2$

$\boxed{f(x) + x\cdot z}$

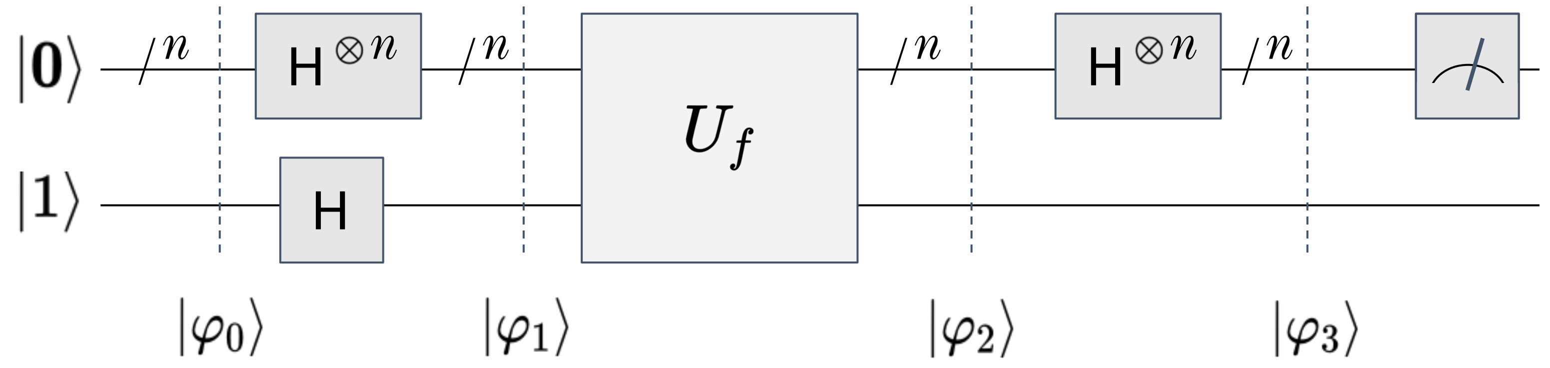
$(s\cdot x + x\cdot z)$

$(s \oplus z)\cdot x$

$$|\varphi_3\rangle = \frac{1}{2^n}\sum_{z\in\{0,1\}^n}\left(\sum_{x\in\{0,1\}^n}(-1)^{(s+z)\cdot x}\right)|z\rangle|-\rangle$$

$|s\rangle$

Bernstein-Vazirani Algorithm



when $z = s$

$$\left(\frac{1}{2^n}\sum_{x\in\{0,1\}^n}(-1)^{(s+s)\cdot x}\right)^2 = \left(\frac{1}{2^n}\sum_{x\in\{0,1\}^n}(-1)^{0\cdot x}\right)^2 = \left(\frac{1}{2^n}\sum_{x\in\{0,1\}^n}1\right)^2 = 1$$

0

$z=s \quad P_s = (\;)^2$

$$|\varphi_3\rangle = |s\rangle|-\rangle$$

Measure

- Measuring the first quantum register yields s with certainty!
- We are able to determines s in 1 oracle query! → n evaluations
- Polynomial (linear) speedup compared to classical algorithm.

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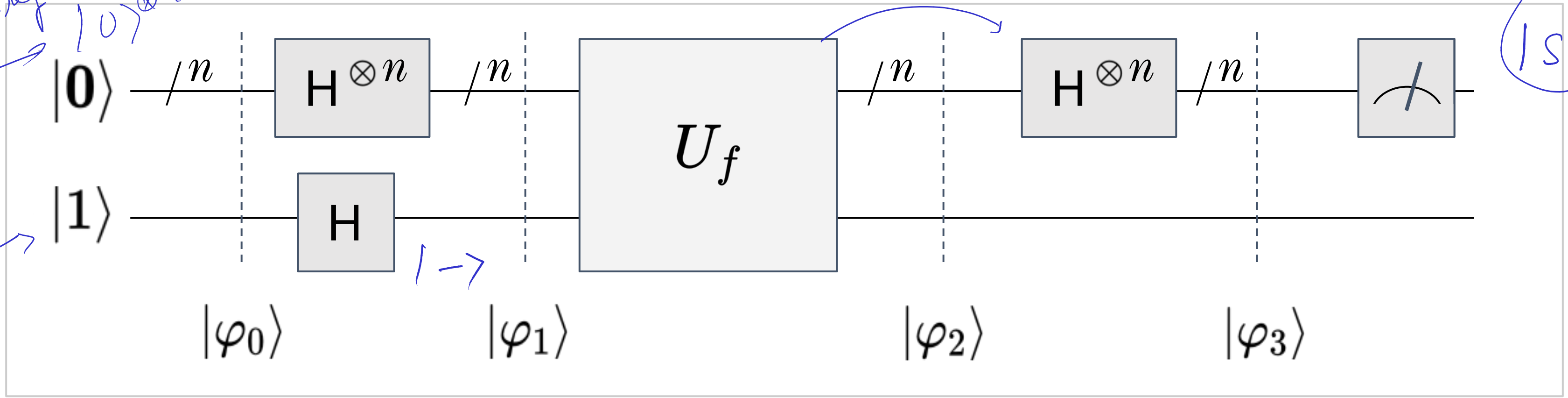
Interpretation via the Deutsch-Jozsa Problem

first reg: $|0\rangle^{\otimes n}$



$|s\rangle$

$\Rightarrow$ exercise : if we have an affine space $(s \cdot x) \bmod 2 + k$, $s \neq 0$

$f(x) = s \cdot x + k$ for $s \neq 0$ ⟶ corresponds to balanced function ← optional exercise.

$f(x) = k$ for $k \in \{0, 1\}$ ⟶ corresponds to constant function

$f(x) \to 0$

$\quad\;\; \to 1$

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> “ Can we use David's idea to construct something even more conspicuous where we can see the separation of power between quantum computation and classical computation? The answer is yes! ”
>
> \- Artur Ekert

Bernstein-Vazirani Algorithm

Creating a framework to build quantum algorithms

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Bernstein-Vazirani Problem

Bernstein-Vazirani Problem

n → 1

$$f : \{0,1\}^n \longrightarrow \{0,1\}$$

unknown

input bitvector

- f is **guaranteed** to be of the form $f(x) = s \cdot x \pmod 2$
- Given a black-box access to f determine the secret string s

$f(\vec{x}_n) = 0$, 1

PROMISE

Classical Solution for generalized problem

$x \longrightarrow n$ bit

$s \longrightarrow n$ bit

- n queries are required to determine s

$x = 1000\ldots0_n$

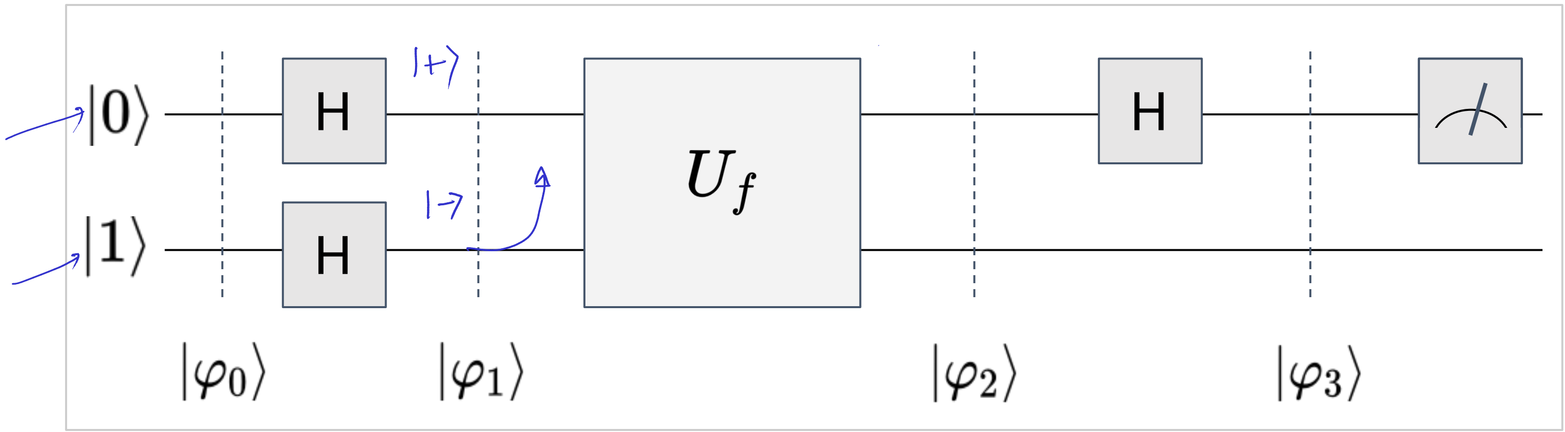
s_1

$$f(100\ldots0_n) = s_1$$
$$f(010\ldots0_n) = s_2$$
$$.$$
$$f(000\ldots1_n) = s_n$$

Quantum algorithm for 1 bit function

1 qubit → 1 bit o/p
n qubits → n bit o/p

Superposition state	Query the oracle	Interfere	Measure



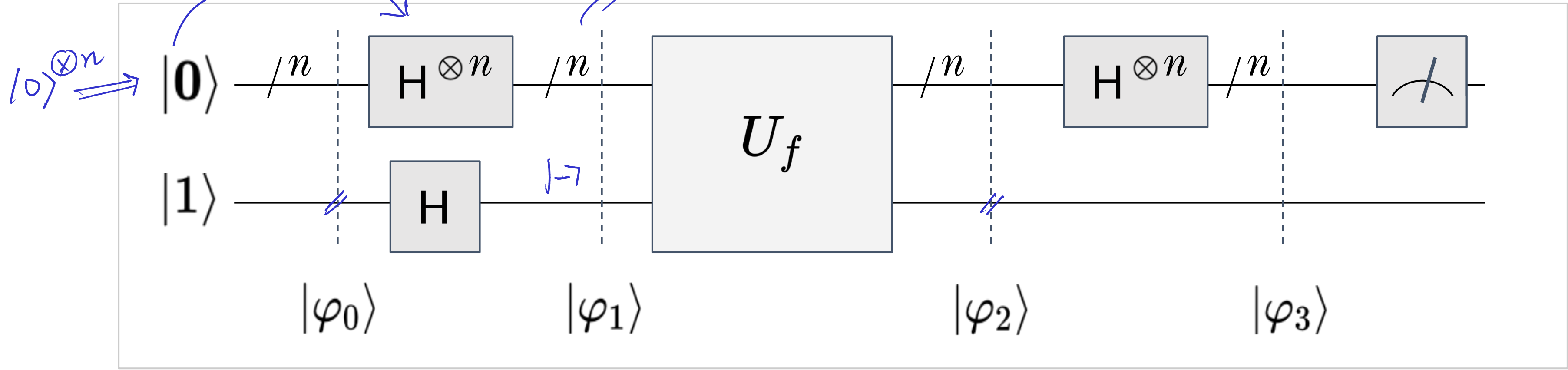
$f(0) \oplus f(1)$

Superposition state	Quantum Parallelism
Ancilla qubit	Phase-kickback
Interference (H)	Extracting information

$|+\rangle \rightarrow |0\rangle$ const
$|-\rangle \rightarrow |1\rangle$ balanced

Deutsch-Jozsa Algorithm

equal superposition state.



$$|x\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

$$(H^{\otimes n} \otimes I)U_f(H^{\otimes n} \otimes H)|\mathbf{0}, 1\rangle$$

$$|\varphi_0\rangle = |\mathbf{0}, 1\rangle$$

$|0\rangle^{\otimes n}$ $(000\ldots0_n) \to (11\ldots1_n)$

$$|\varphi_1\rangle = \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n} |\mathbf{x}\rangle}{\sqrt{2^n}}\right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

$$|\varphi_2\rangle = \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n} (-1)^{f(\mathbf{x})} |\mathbf{x}\rangle}{\sqrt{2^n}}\right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

$f: \{0,1\}^n \to \{0,1\}$

$$\langle , \rangle : \underset{z}{\{0,1\}^n} \times \underset{x}{\{0,1\}^n} \longrightarrow \{0,1\} \equiv {}^0_1$$

$$\langle \mathbf{z}, \mathbf{x} \rangle = \langle z_0 z_1 z_2 \ldots z_{n-1}, x_0 x_1 x_2 \ldots x_{n-1} \rangle$$
$$= (z_0 \wedge x_0) \oplus (z_1 \wedge x_1) \oplus \cdots \oplus (z_{n-1} \wedge x_{n-1})$$

addn mod 2 arithmetic

$|x\rangle|-\rangle \xrightarrow{U_f} (-1)^{f(x)} |x\rangle |-\rangle$

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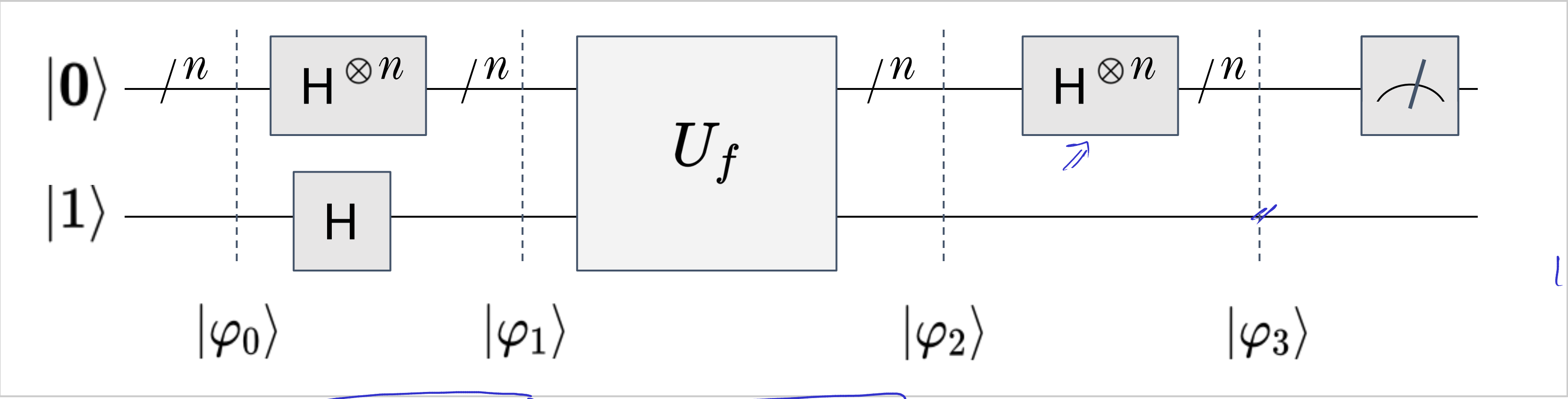
Deutsch-Jozsa Algorithm



$|0\rangle^n \to H^n \to U_f \to H^n$

$|1\rangle \longrightarrow$

$$|\varphi_3\rangle = \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}\sum_{\mathbf{z}\in\{0,1\}^n}(-1)^{\langle\mathbf{z},\mathbf{x}\rangle}|\mathbf{z}\rangle}{2^n}\right]\otimes\left[\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right]$$

exercise

add^n mod

$$|\varphi_3\rangle = \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n}\sum_{\mathbf{z}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}(-1)^{\langle\mathbf{z},\mathbf{x}\rangle}|\mathbf{z}\rangle}{2^n}\right]\otimes\left[\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right]$$

$$= \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n}\sum_{\mathbf{z}\in\{0,1\}^n}(-1)^{f(\mathbf{x})\oplus\langle\mathbf{z},\mathbf{x}\rangle}|\mathbf{z}\rangle}{2^n}\right]\otimes\left[\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right]$$

$$|\varphi_3\rangle = \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}|\mathbf{0}\rangle}{2^n}\right]\otimes\left[\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right]$$

$|0\rangle^{\otimes n}$

$f(x) \oplus \langle z,x\rangle$ add^n mod 2

$f(x) + z\cdot x$

sum in add^n mod 2 arithmetic -

$z = 0$ $\quad \langle 0,x\rangle = 0$

$f(x) \oplus \langle 0,x\rangle = f(x)$

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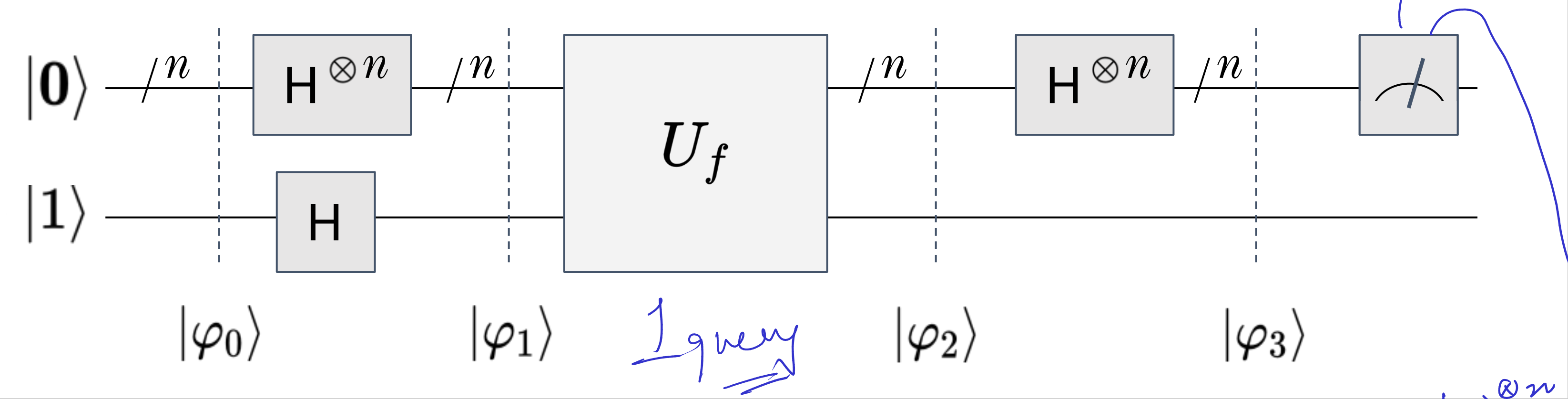
Deutsch-Jozsa Algorithm

$(2^{n-1}+1)$



$|0\rangle \Rightarrow$ constant

$|0\rangle^{\otimes n}$

$f(x)$

$|0\rangle$

any other ⇓ balanced

$\rightarrow |0\rangle^{\otimes n}$

What is the probability that the top qubits will collapse to the state $|\mathbf{0}\rangle$? We can answer this question by setting $\mathbf{z}=\mathbf{0}$ and realizing that $\langle \mathbf{z},\mathbf{x}\rangle = \langle \mathbf{0},\mathbf{x}\rangle = 0$ for all $\mathbf{x}$.

$$|\varphi_3\rangle = \left[\frac{\sum_{\mathbf{x}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}|\mathbf{0}\rangle}{2^n}\right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

The probability of collapsing to $|\mathbf{0}\rangle$ is totally dependent on $f(\mathbf{x})$

$$\frac{\sum_{\mathbf{x}\in\{0,1\}^n}(-1)|\mathbf{0}\rangle}{2^n} = \frac{-(2^n)|\mathbf{0}\rangle}{2^n} = \boxed{-1|\mathbf{0}\rangle} \longrightarrow f(\mathbf{x}) \text{ is constant at } 1$$

constructive interference

$$\frac{\sum_{\mathbf{x}\in\{0,1\}^n}1|\mathbf{0}\rangle}{2^n} = \frac{2^n|\mathbf{0}\rangle}{2^n} = \boxed{+1|\mathbf{0}\rangle} \longrightarrow f(\mathbf{x}) \text{ is constant at } 0$$

destructive interference

$$\frac{\sum_{\mathbf{x}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}|\mathbf{0}\rangle}{2^n} = \frac{0|\mathbf{0}\rangle}{2^n} = 0|\mathbf{0}\rangle \longrightarrow f(\mathbf{x}) \text{ is balanced}$$

$f: \{0,1\}^n \longrightarrow \{0,1\}$

x: n-bit vector
y: 1 bit

" Such a [quantum] computation also need not necessarily evaluate functions when it is solving problems, because the state of its output might be a coherent superposition of states corresponding to different answers, each of which solves the problem. This allows quantum computers to solve problems by methods which are not available to any classical device."

- David Deutsch and Richard Jozsa (1995)

Deutsch-Jozsa Algorithm

Creating a framework to build quantum algorithms

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The generalized case for n bit function

Generalization of Deutsch Problem

$$f : \{0,1\}^n \longrightarrow \{0,1\}$$

- f is **guaranteed** to be balanced or constant
- Given a black-box access to f
- Determine whether f is **balanced** or **constant**

PROMISE

Classical Solution for generalized problem

- Best-case scenario: first two inputs have different outputs assuring that the function f is balanced
- Worst case scenario: $2^{n-1} + 1$ function evaluations
- Classical solution: **deterministic**
- Probabilistic: $P_{\text{constant}}(k) = 1 - \dfrac{1}{2^{k-1}}$ for $1 < k \leq 2^{n-1}$

constant $\Rightarrow$ $\boxed{f(x^*) = 0}$ $\forall x^*$

or $\boxed{f(x^*) = 1}$ $\forall x^*$

half $x^* \Rightarrow 0$

half $x^* \Rightarrow 1$

$x^* \longrightarrow \boxed{U_f} \longrightarrow f(x^*)$

$f(000) = 0$ (0000)

$f(001) = 1$ $\Rightarrow$ balanced.

$f(000) = 0$

$f(001) = 0$

$f(010) = 0$

1 $\Rightarrow$ balanced

1 11 2^n

const ato

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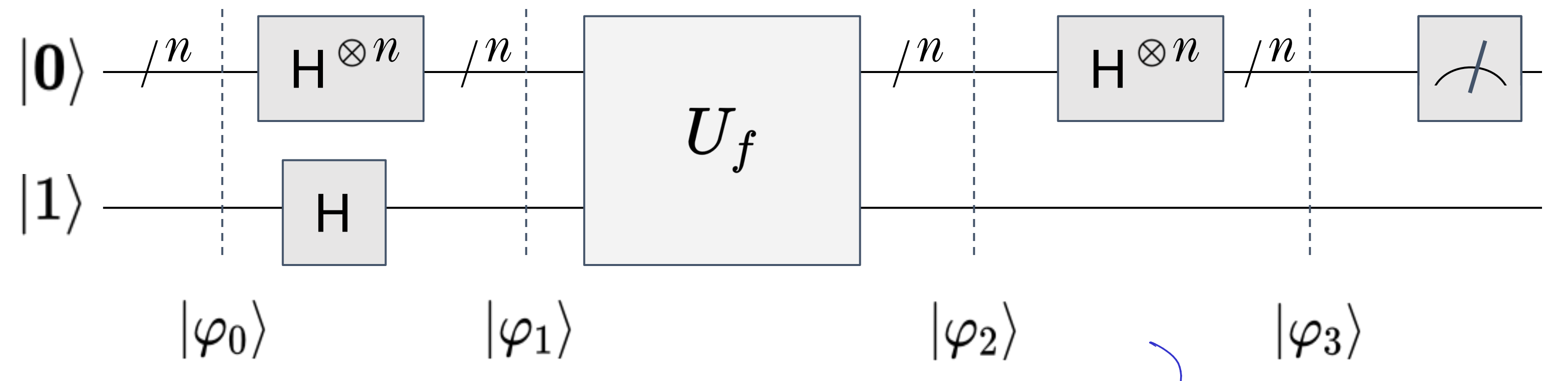
Summary

Superposition state | **Query the oracle** | **Interfere** | **Measure**



Deutsch Algorithm	$f(x) : \{0,1\} \rightarrow \{0,1\}$	**balanced** or **constant**
Deutsch-Jozsa Algorithm	$f(x) : \{0,1\}^n \rightarrow \{0,1\}$	**balanced** or **constant**
Bernstein-Vazirani Algorithm	$f(x) : \{0,1\}^n \rightarrow \{0,1\}$	$f(x) = s \cdot x (\text{mod } 2)$

$|s\rangle$

Three simple algorithms have provided us with a framework to understand quantum algorithm design. It has also described how effects such as superposition, entanglement and interference are used as resources in quantum algorithms. It has also described the phase-kickback trick that is important to understand more complex algorithms like Grover's algorithm, Quantum Fourier Transform, Shor's algorithm.

Some advanced (optional) topics

1985 → 2 queries.

1. Quantum theory, the Church–Turing principle and the universal quantum computer, David Deutsch (https://royalsocietypublishing.org/doi/10.1098/rspa.1985.0070)
2. Rapid solution of problems by quantum computation, David Deutsch and Richard Jozsa (https://royalsocietypublishing.org/doi/10.1098/rspa.1992.0167)
3. Quantum algorithms revisited, R. Cleve, A. Ekert, C. Macchiavello and M. Mosca (https://royalsocietypublishing.org/doi/10.1098/rspa.1998.0164) 1 query
4. Recursive Bernstein-Vazirani Algorithm (https://courses.cs.washington.edu/courses/cse599d/06wi/lecturenotes7.pdf)
5. On the power of quantum computation (https://ieeexplore.ieee.org/document/365701)

✓ Simon Problem] ⇒

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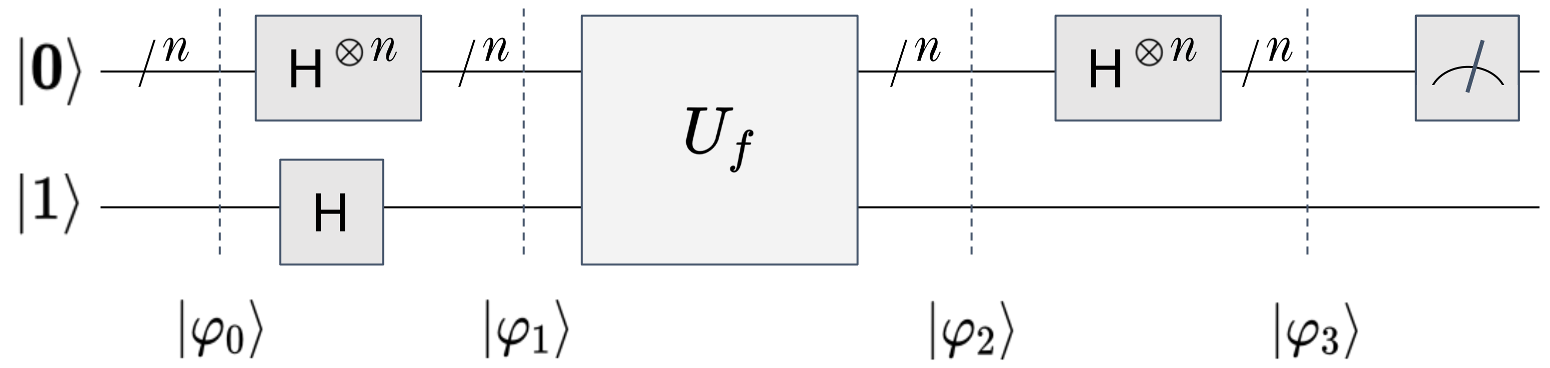
Deutsch-Jozsa Algorithm

$f: \{0,1\}^n \longrightarrow \{0,1\}$



- Problem: Determine whether function is balanced or constant (promise problem)
- Classical algorithm: $2^{n-1} + 1$ queries (function evaluations)
- Deutsch-Jozsa algorithm: 1 query (function evaluation), with 100% **certainty**
- Quantum algorithm provides an exponential advantage in terms of query complexity.

$|0\rangle^{\otimes n} \Rightarrow$ const

$\Rightarrow$ balanced

Optional Exercise:

Write a probabilistic classical algorithm for solving the generalized Deutsch Problem and determine whether it can guarantee correctness. This exercise should provoke your thoughts about classical algorithms and quantum algorithms!